LOT COMPUTER ��? PLAN

LOT COMPUTER — Hardware Plan, Feasibility & Roadmap

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DOCUMENT   LOT-COMPUTER / 01 PLAN
ISSUE      2026-10-11            REV A
CLASS      INTERNAL / ENGINEERING
S-2        VADIK MARMELADOV
STATUS     PLAN — nothing built, nothing ordered
SIBLINGS   02 BOM · 03 FIRMWARE · 04 API CONNECTOR · 05 HOST SOFTWARE · 06 MANUAL
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00 PRINCIPLE

A pocket object that receives **pager-style autonomous notifications** from lot-systems.com (“Coffee time!”), lets the holder press one button (“Copy”) to write that moment into the site **Log tab**, and reports weather / air data back. Same LOT rule as NODE-0: input → classify → action → record. The device never acts silently; every press and every reading leaves a Log entry.

01 REQUIREMENT TRACE (the 19 items)

#	Requirement	Where it lands	Status
1	PCBWay	PCB fab + assembly + CNC steel (BOM §A, §F)	PLANNED — quote needed
2	Pager-like notification from AI site	API doc §3 `GET /device/notifications`, firmware notify state	SPEC’D
3	2-part stainless body	Back plate + front bezel, 316L (BOM §F)	SPEC’D — see D-1
4	Flat silver square 40×40×5 mm	v1 = 40×40×9 mm; 5 mm = v2 gate	**RISK R-1**
5	Camera	OV2640 (v1) → thinner module (v2) (BOM §D)	OPEN — purpose undefined, Q-1
6	LOT API connector	API doc	SPEC’D — server not built
7	PDF manuals	Doc 06 + `pdf/` build	DONE (draft PDFs)
8	Compress info per session	API doc §5 session digest	SPEC’D
9	Firmware documents	Doc 03	DONE (draft)
10	Software to connect to firmware	Doc 05 `lotctl`	SPEC’D
11	Separate documents	docs 01–06 are separate files	DONE
12	Charger	Qi pad (BOM §E)	SPEC’D
13	100-unit run	Roadmap P6, BOM totals	PLANNED
14	Weather sensor	BME688 + SHT45 (BOM §C)	SPEC’D
15	AI-grade off-the-shelf sensors	BME688 gas scanner (trainable) + ESP32-S3 TinyML	SPEC’D
16	“Copy” button → Log tab	API doc §4, uses existing `POST /api/logs`	SPEC’D
17	One side polished stainless	Back plate, mirror polish	SPEC’D — see D-1
18	Other side camera + screen + button	Front bezel + glass	SPEC’D
19	Wireless charging	Qi receiver (BOM §E)	SPEC’D — see D-1

02 BLOCK DIAGRAM

```
lot-systems.com (Fastify, /api)
  │  HTTPS (TLS) · device token
  ▼
[Wi-Fi] ESP32-S3 ──I2C── BME688 (T/RH/P/gas) · SHT45 (T/RH) · VEML7700 (light)
  │  │  └─DVP── camera
  │  └─SPI─── display (square, ~1.3")
```

```
   │  └GPIO── button "Copy" · status LED (camera-active)
   └I2C── fuel gauge ── charger IC ── LiPo ◄── Qi receiver IC ◄── Rx coil
```

03 FEASIBILITY ANALYSIS (read this before buying anything)

R-1 — 5 mm thickness is the hardest requirement. Rough stack, ESTIMATE:

Layer	**mm**
Steel back plate floor	0.6
Battery (thin LiPo)	2.0 (≈100–150 mAh)
PCB (1.0 incl. parts)	1.0
Display + cover glass	1.2
Steel front bezel lip	0.4
Total	**≈5.2**

That is before the camera, whose module alone is typically thicker than 4 mm (off-the-shelf OV2640 modules). A packaged ESP32-S3 module is 3.2 mm tall, so a 5 mm build also needs the bare ESP32-S3 chip. Conclusion: **v1 (EVT/DVT) is 40×40×9 mm** with dev-friendly parts; **5 mm is a v2 gate** requiring a chip-down design, a micro camera (candidates: OmniVision OVM-series cube sensors; not verified), and a ≤100 mAh battery. Battery life at 5 mm will be days, not weeks, with Wi-Fi polling.

R-2 — Stainless steel blocks radio and Qi. Steel is a conductor: Wi-Fi/BLE antennas and a Qi coil cannot sit behind it. Decision **D-1** (needs S-2):

- **Option B (recommended):** polished back plate with a ~16 mm non-metal inlay (zirconia ceramic or glass) over the Qi coil; antenna under the front glass edge, in a non-metal slot of the bezel. Preserves “one side polished steel” visually; the inlay can carry the LOT mark.
- Option A: coil + antenna under front glass only; device must be charged face-down on the pad (screen covered). Simplest, worst UX.
- Option C: full-steel back, contact (pogo) charging. Rejects req. 19.

R-3 — Certification. A pre-certified ESP32-S3 module keeps FCC/CE effort low; a chip-down board needs its own radio testing (ESTIMATE $8–20k). Lithium cells need UN38.3 transport documentation. A 100-unit *sale* needs both; 100 units given to staff/testers less so (verify with a compliance lab).

R-4 — Camera privacy. A camera on a notification device needs a hardware indicator LED tied to the sensor power rail, capture only on explicit press, and clear opt-in on the site. Camera *purpose* is undefined (Q-1).

R-5 — Third-party licenses. BME688 on-chip AI (Bosch BSEC2 / BME AI-Studio) uses Bosch’s proprietary library; licence terms for a commercial product must be read before the 100-unit run (not verified in research).

R-6 — Supply. DigiKey listed 8–12 week lead times for BME688 and ESP32-S3 modules, one regional listing showed zero stock. Order samples now.

04 DECISIONS & OPEN QUESTIONS FOR S-2

- **D-1** Back-side inlay for Qi + antenna (Option B recommended).
- **D-2** v1 at 9 mm, 5 mm as v2 — accept? (recommended: yes).
- **Q-1** What does the camera do? Assumed: still photo on double-press, uploaded as a Log attachment. Needs confirmation.
- **Q-2** Is the 100-unit run for sale (needs certification) or internal?
- **Q-3** Notification delivery: device polls (assumed, battery-cheap) vs. keeps an SSE/WebSocket open (instant, costs battery).
- **Q-4** The brand sites (brand.lot-systems.com, lot-systems.com/about, institute.lot-systems.com/cqgs.html) were **not reachable** from this environment (DNS failure). Brand colour/type rules were taken from `docs/technical/LOT-STYLE-GUIDE.md` and `LOT-NODE-0-RIG-SPEC.md` only; CQGS was not read. Re-check on a networked session.

05 ROADMAP

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PHASE                        WEEKS   EXIT GATE
P0 Spec lock (D-1,D-2,Q-1..3)  0–2    S-2 signs decisions
P1 Dev-kit prototype           2–7    DFR1221 + sensors: notify→Copy→Log works end-to-end
P1b Server: device API         2–7    endpoints + migration live on staging (parallel)
P2 EVT PCB (PCBWay)            6–14   board boots, all sensors read, Qi charges
P3 Enclosure samples (PCBWay)  8–14   2-part steel fit check, polish approved
P4 DVT, 10–20 units            14–24  soak test, drop test, battery life, thickness verdict
P5 Certification               20–30  FCC/CE/UN38.3 reports (if for sale)
P6 PVT → 100-unit run          30–38  100 units pass end-of-line test
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Total ≈ 9 months; critical path = P3→P4→P5 (steel finishing, thickness, labs). Roadmap is ESTIMATE — dominated by component lead times (8–12 wk) and lab queues.

06 BUDGET (ESTIMATE)

See BOM doc: ≈ **$120–220 per unit** → **$12–22k** for 100 units, plus NRE (fixtures, prototype rounds, certification) ≈ **$15–35k**. All figures are ESTIMATE ranges, not quotes.

07 NEXT ACTIONS

1. S-2 answers D-1, D-2, Q-1..Q-4.
2. Order dev kit + samples (BOM §Prototype).
3. Request PCBWay quotes: PCB+assembly (4-layer, ~40×40 mm) and CNC 316L ×2 parts, mirror polish, qty 5 and 100.
4. Build server device endpoints (API doc), staging first.
5. Firmware milestone M1 on dev kit.